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AXAN-Net: An Adaptive Exponential Approximator Neural Network for Latent Feature Composition with Application in Audio-Based Fan Fault Detection

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***Abstract*— This paper introduces AXAN-Net, a neural architecture which integrates a Fully Connected Nonlinear Network (FCNN) with an Adaptive Exponential Feature Network (AEFN). The FCNN uses a gating mechanism to assign feature-wise importance scores, producing a modulated input vector that highlights informative components of the audio signal. This gated representation is then passed to the AEFN, where learnable exponential units with trainable centers and deviations transform the features into a smooth nonlinear space. The final output is projected onto a compact two-dimensional latent space, enabling a simple linear classifier to separate operating states with high accuracy. The model was evaluated using acoustic data recorded from a ventilation fan operating under three conditions: healthy, unbalanced type 1, and unbalanced type 2. AXAN-Net achieved a macro F1-score of 97% on the training set and 95% on the test set, outperforming PCA, SVD, MLP, RBF, and CNN-based methods. The resulting 2D embedding provided clear visual class separation and showed improved robustness under Min-Max scaling. The implementation of AXAN-Net is publicly available on GitHub.**

Keywords*— *Feature Extraction; Fault Detection; Condition Monitoring; Deep Learning.

I. Introduction

Reliable fault diagnosis is central to the safe and efficient operation of industrial machines such as ventilation fans. Undetected faults can cause vibration, noise, energy loss, and costly downtime [1], while overly conservative maintenance schedules waste resources [2]. Modern condition monitoring addresses this trade-off by continuously assessing equipment health and initiating maintenance only when necessary [3]. Among sensing modalities, audio and vibration–derived features have emerged as practical signals for early fault signatures [4], enabling non-destructive monitoring without disrupting production. Concurrently, Machine Learning (ML) has transformed fault diagnosis by learning discriminative patterns directly from data. However, key challenges still prevent the widespread adoption of ML models in real industrial settings. Cost and latency cause models which are large, complex, or require extensive feature engineering slow in decision-making and inflate inference costs; Mean whole data volume in industrial datasets can be high-dimensional and redundant, increasing storage, transmission, and training overheads while risking overfitting [5]. Effective solutions must therefore deliver high diagnostic accuracy with low computational costs, and compress large feature sets without sacrificing class separability or accuracy [6]. What is more, while MinMax scaling to $[0, 1]$ range is frequently used in machine learning, in auditory data, local amplitude variations and noise cause reduction in performance of common machine learning methods such as MLPs and CNNs and need for better network design is needed under this type of normalization [7]. To address these needs, this paper proposes AXAN-Net, an Adaptive Exponential Approximator which couples a gated

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Non-Linear Fully Connected Network (FCNN) for feature-wise selection with a nonlinear exponential kernel transformation and a compact 2-D latent space for classification. The gated FCNN emphasizes informative audio features and scale the importance of each input feature, while the kernel layers capture nonlinear relations essential for distinguishing healthy and unbalanced fan states [8] so that linear classifier can be used to separate the classes. By projecting from a 10-dimensional input to a 2-D latent space, AXAN-Net reduces data size while maximizing the separation of different classes, enabling low detection error rates and high accuracy while managing the depth and number of parameters, which are key advantages for a real-time condition monitoring [3]. Lastly, in order to compare the current model performance with machine learning methods within the literature, comparative and preprocessing methods similar to [9] was utilized to establish performance of the model. Moreover, to quantify how well the model performed in terms of separation in the latent space, clustering metrics from [10] where used. Last but not least, it is shown that under Min-Max normalization, AXAN-Net performs significantly better than classic machine leaning methods.

II. DATA ACQUISITION

A. *Setup and Data Collection*

In this study, three distinct experimental conditions were established: (1) a healthy ventilation fan, (2) an unbalanced fan of type 1, and (3) an unbalanced fan of type 2. To generate the unbalanced conditions, a mass weighted 5 grams was affixed to one of the fan blades, as showed in Figures 1. Acoustic data were captured using a microphone positioned at a fixed distance of 10 centimeters from the fan. Each condition was recorded for approximately 11 minutes, resulting in a total recording duration of 33 minutes across all experiments. The collected audio was subsequently divided into 2000 segments of roughly 1 second each, forming the basis of the raw dataset used for analysis. The experimental data were collected in our previous study in [11].

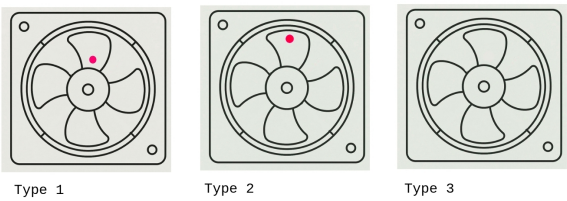

Figure 1. Fault type 1, fault Type2 and Type 3(Healthy)

B. *Data Preprocessing*

The dataset used in this study consists of 2000 samples. Each sample represents a set of features extracted from the corresponding audio signal. The raw dataset, stored in Excel format, was loaded and pre-processed prior to model training. All feature columns were converted to numeric values, with non-numeric entries replaced by "Nan." Missing values were imputed using the column-wise median. The dataset was then divided into 80% training and 20% testing subsets using stratified sampling to preserve class distribution. Finally, each feature was standardized using Min-Max normalization, where normalization was fitted to the training subset and was applied to testing subset to avoid data leakage and realistic comparison of performances. Figure 1 presents the data statistical histogram for the normalized training and testing subsets. As it can be seen in Figure 1, although the distributions of train and test are similar, there are many samples in which the feature values or feature values differs, which indicates data leakage was kept to a minimum.

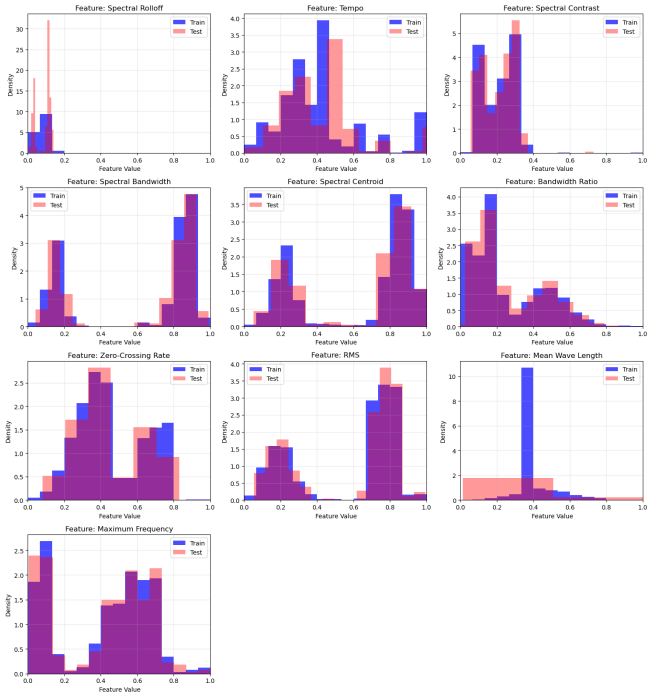

Figure 2: Training and testing features histogram plot with 15 bins per feature. As it can be seen, the distributions are not identical and suggests that testing subset can be used as a comparison for overfitting measure.

III. AXAN-NET FOR 2-D FEATURE EXTRACTION AND CLASSIFICATION

In this section, AXAN-Net is introduced as a neural architecture designed to adaptively approximate latent feature compositions from audio signals for fan fault detection. The method is structured by integrating a fully connected non-linear network with a nonlinear exponential kernel transformation and a low-dimensional latent projection using linear classifier, through which feature importance is shown to be learned and nonlinear relationships are shown to be captured. In this section, first, the layers and Model architecture are explained, after which, training and evaluation processes are described.

A. *AXAN-Net Layers and Architecture*

The proposed model combines a gated Fully Connected Non-linear Network (FCNN) with nonlinear exponential kernel transformation, projecting input features into a 2-dimensional latent space for classification. The architecture is described in Figure 3, along with the activation functions and weight matrix sizes. As it can be seen from Figure 3, the FCNN selects the most important features through the non-linear mechanism and then multiplies it to the original feature to get a weighted effect of each feature, which is

then composed using the exponential layer, which in essence, is estimating the best way to compose these modulated features for the end of best separation using the line regression and achieving lowest classification loss. Moreover, the activation functions used in the FCNN are proportional to feature values, meaning that saturation effects and gradient vanishing is reduced during the training. Moreover, the dynamic and input dependent structure of the model prompts the ability to generalize to a non-IID test domain and avoid overfitting, especially in cases where data points of different classes are mixed in between each other distingue able class clusters that results in high accuracy cannot be formed by classical machine learning methods.

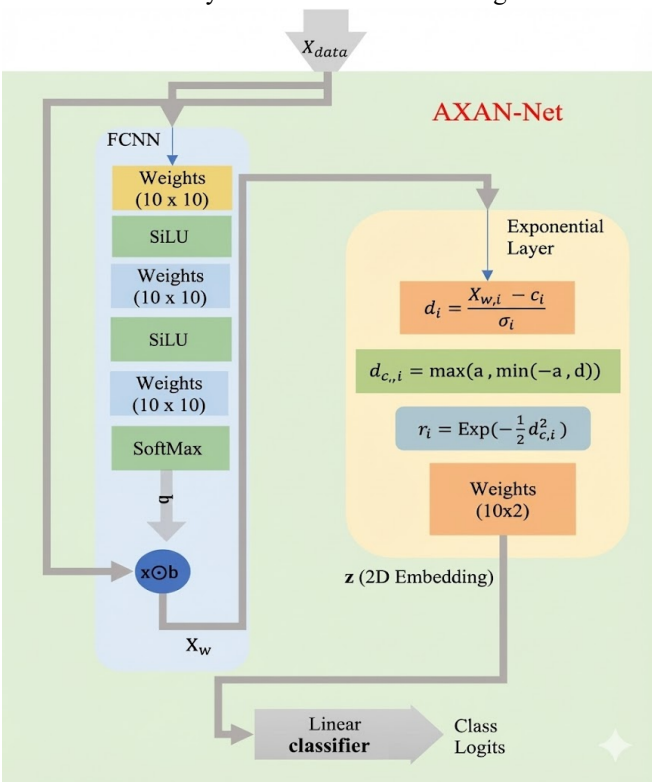

Figure 3. Model Architecture. shows how the data is first selected by the FCNN and then how it is composed by the exponential layer to get the 2-D embedding, which in turn is used in the linear classifier to get the class logits.

The rest of this part discuss the layers utilized in AXAN-Net's architecture.

1) Fully Connected Non-Linear Network Layer

As it can be seen from Figure 3, the input feature vector $X_d \in R^d$ is first passed through the FCNN. This layer consists of two weighted matrixes separated by a Sigmoid Linear Unit (SiLU) activation and a weight matrix which output of it is activated by a Softmax. The formulation for these to activations are shown below. Here x_i is the i-th element of the input vector to the activations and $SiLU(x)_i$. $SoftMax(x)_i$ are the i-th vector output of the activations.

$$SiLU(x)_i = x_i \times \frac{1}{1+\exp(-x_i)} \quad . \quad SoftMax(x)_i = \frac{\exp(x_i)}{\sum \exp(x_i)} \qquad (1)$$

The output of the Softmax is then is modulated by the original input via element-wise multiplication to make the x_{weighted}. This gating mechanism allows the network to selectively emphasize informative features and suppress less relevant ones. It is important to note that a SiLU activates each feature based on the value of that said feature.

2) Exponential Layer

The weighted features are then processed through the Exponential layer, which introduces non-linear feature transformations. Each input dimension has a learnable center c_i and deviation σ_i. The activation is computed as:

$$z_i = \exp\left(-\frac{1}{2} \times \left(\frac{x_{\text{weighted}.i} - c_i}{\sigma_i}\right)^2\right) \qquad (2)$$

To manage z_i values and avoid overflow, a clipping is done on the distance value (d) which is equal to $\frac{x_{\text{weighted}.i} - c_i}{\sigma_i}$ .The clipped value of d was defined as $d_c \in [-a, +a]$, $a > 0$ resulting in computationally manageable vector $z \in R^d$ which represents a non-linear embedding of the input. Finally, z vector is passed through another set of learnable weights to get the 2D embedded version of z vector.

3) Classification Layer

Finally, the 2D latent vector is passed through a linear classifier to produce logits for the target classes. Then during the training and testing, the cross-entropy loss applies an internal soft-max to these logits and penalizes negative log-likelihood of the mismatches through the optimizer. Moreover, KLD-Loss was used between 2D embedding and target labels to provide enhanced separation context for 2D embedding representation.

B. Training and Evaluation Procedures

The proposed model was trained using a supervised learning approach with mini-batch gradient optimization. All experiments were conducted on a workstation equipped with an NVIDIA GPU. Moreover, simultaneous training was done on a MacBook equipped with MPS GPU to ensure reproducibility. The training process utilized the Adaptive Moment Estimation (Adam) optimizer with a learning rate of 1e-4 with a batch size of 64 and weight decay of 1e-5. The model was trained for 100 epochs. The Cross-Entropy Loss function was employed to measure the mismatch between the predicted class probabilities and the true labels, while KDL-Loss was utilized for the 2D embedding. During each training epoch, the input samples were randomly shuffled to ensure robust generalization. For each mini-batch, the model performed a forward pass to compute the logits, latent representations (z), and exponential layer activations characterized by their centers and deviations. The loss gradients were then back-propagated through the network, and the model parameters were updated using the Adam optimization algorithm. To stabilize training, clipping with maximum norm of 500 was applied. After each batch update, performance metrics such as the macro F1-score were computed to monitor the model's learning behavior. Additionally, the latent feature representations and the evolution of exponential centers and deviations were visualized to provide insights into the model during training. Validation performance was periodically evaluated using the 10 percent held-out validation set of training subset to assess generalization capability before testing.

The proposed model was evaluated on test dataset to evaluate generalization and prevent overfitting. Performance

was measured at each epoch using the F1-score, a balanced metric suitable for multi-class tasks. During evaluation, the model operated in inference mode to disable gradient updates. Predictions and cross-entropy losses were computed to assess accuracy and convergence. Additionally, latent feature visualizations were analyzed to explore class separation.

Moreover, for testing comparisons, a 6 layer Deep Simple MLP with batch normalization and hidden neuron size of 10, an RBF network with 10 centers and a linear projection layer at the head, a CNN with two convolutional blocks and batch normalization with 10 feature maps per each layer with a 2 layer MLP head, PCA with a KNN head and finally a SVD with a KNN head were used against AXAN-Net and all methods were trained for 100 epochs with the training subset and were tested on metrics with the testing subset. Lastly, all supervised methods that were compared with AXAN-Net used batch normalization in their respective latent space for enhanced visualization and training stability. Moreover, dropout was used in CNN and MLP to encourage more stable training and avoid possible overfitting. All measures such as using deeper MLPs and small kernel sizes with paddings and having similar number of parameters were taken so that competing methods to AXAN-Net have a fair comparison.

IV. Experimental Results

In this section, the obtained results are presented and analyzed. The proposed model's performance is compared with baseline methods from both quantitative and qualitative perspectives. Quantitative analyses include numerical evaluation metrics such as F1-score, and loss, while analyses focus on visual interpretations of latent feature spaces, RBF activations, and class separability patterns.

A. F1-Score

After training for 100 epochs the network achieved a macro-F1 score of 0.9530 on the validation set and 0.9727 on the training data, while the testing showed macro-F1 score of 0.9554 for 100 epoch of training. These results indicate, firstly, that the model performs very well on the dataset, and secondly, that no signs of overfitting are observed. In fact, the model reached a stable performance after approximately 70 epochs, but the training was continued for additional epochs as shown in Figure 4.

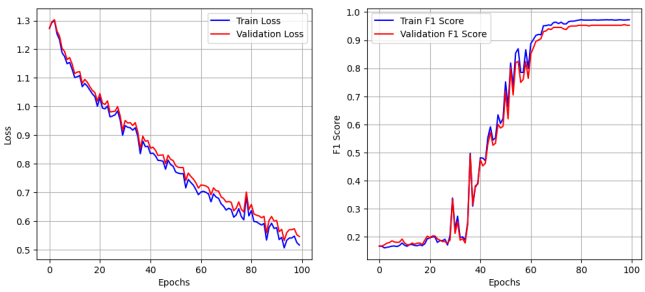

Figure 4. F1 score and losses through epochs for training and train hold out set (validation).

B. Exponential Layer Parameters

The evolution of the Exponential layer parameters during training is shown in Figures 3 and 4, which show the progression of the trainable centers and deviations across epochs. As shown training proceeds, the centers gradually shifted and stabilized in the [-0.3, 0.1] range. Similarly, σ values converged to the [2 , 4] range. These trends demonstrate that the exponential layer refines its parameters to capture nonlinear feature dependencies and enhance classification performance seen in training and testing accuracy and F1.

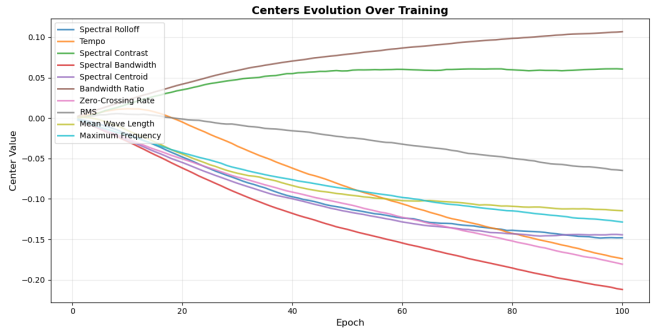

Figure 5. Evolution of Centers each selected features across training epochs.

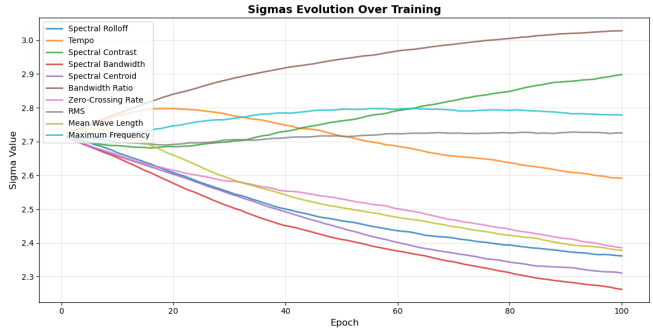

Figure 6. Evolution of σ of each selected features across training epochs.

C. Comparison of the Results with Classical Methods and 2D latent space comparison

To evaluate the quality of the learned feature representations, several two-dimensional latent layers of machine learning methods were applied to the original data and were compared to latent space of the proposed model, as shown in. The projections obtained by PCA, SVD, CNN, MLP and RBF reveals compact clustering patterns did not occur in the 2D embedding for the purpose of correct classification and decrease of mismatch error. These Latent space representations can be seen for PCA in Figure 7, for MLP in Figure 8, for RBF in Figure 9 and For CNN in Figure 10. Meanwhile, Corresponding latent space of AXAN-Net, seen in Figure 11 shows separation and clustering of three classes into three distinguished groups, confirming that the network successfully separates different operational states in a low-dimensional space, enabling the linear classifier to separate the classes with ease. Among competing methods, the PCA visualization comes second and exhibit the most amount of separation of Class 1 in comparison to other classes, demonstrating the nonlinear separation towards correct class prediction, captured by AXAN-Net in comparison. What is more, although MLP reaches good separation of data points, this clustering is artificial and classes 2 and 3 cannot be distinguished from

each other by MLP. Moreover, the found separation in 2D latent space for both CNN and MLP although compact, are distributed on the two normal axis and are not trivially separated in the last layer. In contrast, AXAN-Net with the help of KDL-Loss archives enhanced latent space separation of data point, while due to the cross-entropy loss, this separation is aimed at the minimizing the linear classifiers error, which is reflected in the high F1-Score and Accuracy. What is more, although RBF did separate class 1 from the 2 other classes in a separated cluster, as seen the possibility of separating class 1 by PCA as an unsupervised method, classes 2 and 3 are not distinguished and predicted as the same class by RBF.

Moreover, Table 1 demonstrates that both F1 score and are better for the proposed model in this work. As it can be seen, the best competing model is the combination of CNN with MLP head, which lacks behind by about 60 percent in accuracy. Moreover, AXAN-Net reaches the best F1-Accuracy. The Deep MLP reaches a higher silhouette score in terms of separating classes in the 2D latent space. However, this separation in MLP's case does not mean that the last layer can simply draw lines to separate classes. On the contrary, AXAN-Net reaches the second highest silhouette score, and here, the better clustering enables the last layers linear classifier to separate classes to the near perfection, excluding the few hard-to-cluster in-between samples, which is reflected in the F1-score and Accuracy of test set. The same trend can be see in the Davies-Bouldin score, where lower score means better separation of classes.

TABLE 1- Metric comparison of test data

Model	F1 Score	Accuracy	Silhouette[10]	Davis -Bouldin [10]
Deep Simple MLP	0.5351	0.6427	0.8158	0.2318
RBF	0.5046	0.6303	0.3479	1.4031
PCA + KNN	0.2518	0.3226	0.3432	1.3212
SVD + KNN	0.2573	0.3176	0.3503	1.8197
CNN + MLP head	0.5346	0.6427	0.3883	1.6744
AXAN-Net	**0.9554**	**0.9553**	0.7163	0.3767

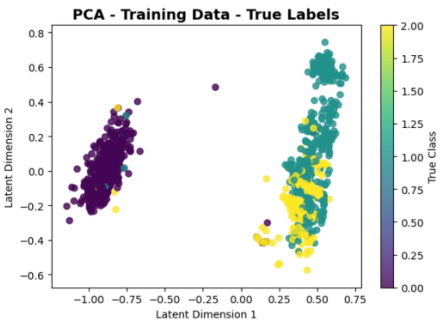

Figure 7. PCA Latent space of the training data to show how separable classes are using unsupervised methods

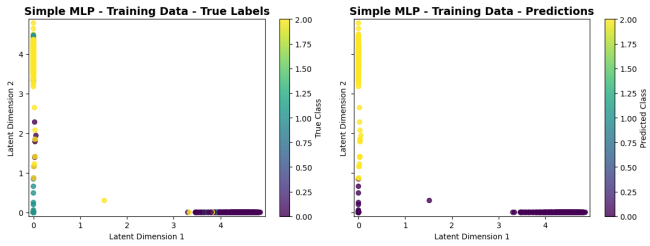

Figure 8. True and Predicted labels for Deep Simple MLP 2D Latent Space

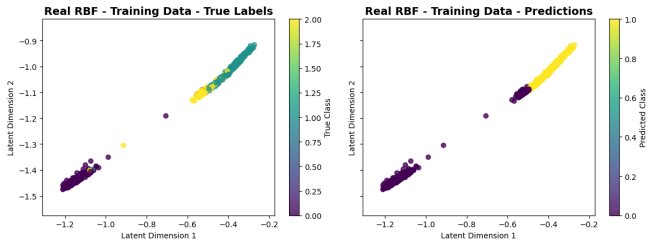

Figure 9. True and Predicted labels for RBF's 2D Latent Space

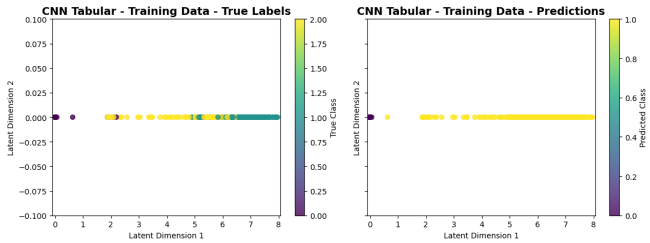

Figure 10. True and Predicted labels for CNN+MLP head 2D Latent Space

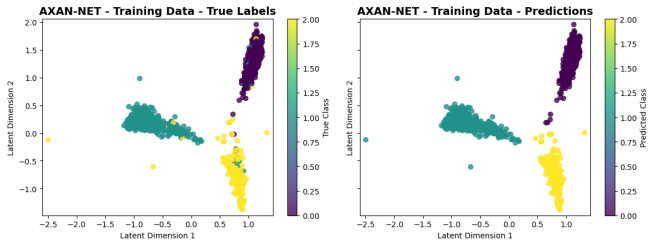

Figure 11. True and Predicted labels for AXAN-Net's 2D Latent Space

V. *Discussion and Conclusion*

This study presented a hybrid neural architecture for ventilation fan fault detection that integrates a gated Fully Connected Network with a nonlinear kernel layer. By combining feature-wise gating and adaptive kernel transformations, the proposed model effectively captured complex relationships in acoustic features while reducing the input space from ten to two latent dimensions. The model achieved high diagnostic accuracy with a macro F1-score of 0.95 on the test set, confirming its strong generalization and robustness against overfitting. Visualization of the latent feature space and kernel parameters demonstrated clear class separability and feature selection. What is more, as expected from [7], while MLPs and CNNs experience loss of performance in signal class processing under Min-Max scaling, the adaptive exponential layer and the gating mechanisms in AXAN-Net compensate for the normalization effects and archives better separation with context of accurate class prediction.

However, AXAN-Net, although as a light weight network in comparison to state of the art models such as multi-input CNNs in [9] or Self-Attention based deep

learning models in, is not yet tested on benchmark datasets or language modeling tasks and further research is needed for this proposed architecture. Moreover, the effectiveness and the methodology to deepen and generalize the layers of this network is still unclear and remains an open problem. What is more, as the dataset used in this work was 1D signals, the performance of AXAN-Net in 2D and 3D data is not evaluated and research is needed in this area as well.

Overall, the approach provides a reliable, efficient, and accurate framework for both fault detection and condition monitoring in ventilation fan systems when using MinMax normalization, offering potential for broader application in industrial monitoring tasks. Last by not least, the presented method in this work contributes a way for non-linear dimensionality reduction, prompting the need for research in the intersection of representation learning and condition and fault monitoring.

REFERENCES

[1] K. Almutairi, H. Wen, and J. K. Sinha, "Standardisation of Vibration-based Parameters for Rotor and Bearing for Machine Faults Detection Using Machine Learning Model," *Journal of Vibration Engineering & Technologies*, vol. 13, no. 7, Sep. 2025, doi: https://doi.org/10.1007/s42417-025-02041-y.

[2] T. A. Al Ameeri, M. N. Ab Rahman, and N. Muhamad, "Analysing Effective and Ineffective Impacts of Maintenance Strategies on Electric Power Plants: A Comprehensive Approach," *Energies*, vol. 16, no. 17, p. 6243, Jan. 2023, doi: https://doi.org/10.3390/en16176243.

[3] O. Surucu, S. A. Gadsden, and J. Yawney, "Condition Monitoring using Machine Learning: A Review of Theory, Applications, and Recent Advances," *Expert Systems with Applications*, vol. 221, p. 119738, Jul. 2023, doi: https://doi.org/10.1016/j.eswa.2023.119738.

[4] L. Yu, X. Yao, J. Yang, and C. Li, "Gear Fault Diagnosis through Vibration and Acoustic Signal Combination Based on Convolutional Neural Network," *Information*, vol. 11, no. 5, pp. 266–266, May 2020, doi: https://doi.org/10.3390/info11050266.

[5] Paleyes, A., Urma, R.-G., & Lawrence, N. D. (2023). Challenges in Deploying Machine Learning: A Survey of Case Studies. https://doi.org/10.17863/CAM.83936.

[6] P. V. Dantas, W. Sabino da Silva, L. C. Cordeiro, and C. B. Carvalho, "A comprehensive review of model compression techniques in machine learning," *Applied Intelligence*, vol. 54, no. 22, pp. 11804–11844, Sep. 2024, doi: https://doi.org/10.1007/s10489-024-05747-w.

[7] L. B. V. de Amorim, G. D. C. Cavalcanti, and R. M. O. Cruz, "The choice of scaling technique matters for classification performance," *Applied Soft Computing*, vol. 133, p. 109924, Jan. 2023, doi: https://doi.org/10.1016/j.asoc.2022.109924.

[8] D. Gao, G. Yang, "Adaptive RBF neural networks for pattern classifications", Proceedings of the 2002 International Joint Conference on Neural Networks (IJCNN), 846–851 (2002). https://doi.org/10.1109/IJCNN.2002.1005584

[9] Ehsan Forootan, H. Ghasemi, Mehdi Tale Masouleh, A. Kalhor, and B. Moshiri, "JoTyMo: Joint Type and Movement Detection from RGB Images of Before and After Interacting with Articulated Objects," pp. 592–597, Dec. 2023, doi: https://doi.org/10.1109/icrom60803.2023.10412492.

[10] F. Ros, R. Riad, and S. Guillaume, "PDBI: A partitioning Davies-Bouldin index for clustering evaluation," *Neurocomputing*, vol. 528, pp. 178–199, Apr. 2023, doi: https://doi.org/10.1016/j.neucom.2023.01.043.

[11] M. Pashaei, A. Aghdaee, M. Khorramabadi Arani, "Audio-based fault diagnosis of ventilation fans using machine learning algorithms", Proceedings of the 14th International Conference on Acoustics & Vibration (ISAV2024), Karaj, Iran, December 2024

[12] Y. Tang, P. Ma, L. Li, X. Liu, Y. Liu, and Q. Wang, "Incipient fault detection based on double Kullback–Leibler divergence KLDattW improved by a self-attention mechanism," *Reliability Engineering & System Safety*, vol. 264, pp. 111247–111247, May 2025, doi: https://doi.org/10.1016/j.ress.2025.111247.